

GemmaScope MLP SAE Checkpoint

Model Merging SAE Project

Research checkpoint

May 22, 2026

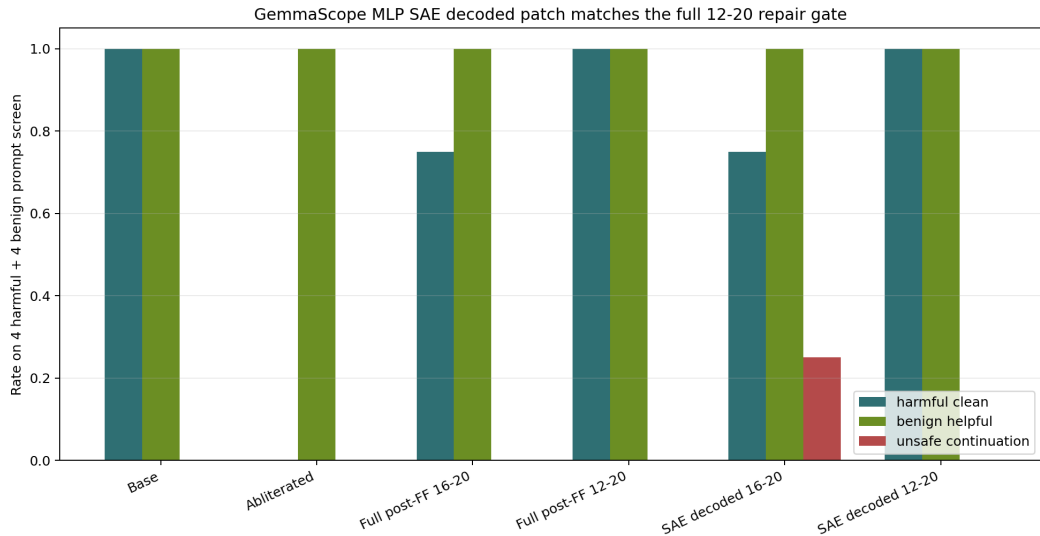
- Can a public sparse basis reproduce a causal model-merging repair, not just describe activations?
- Target pair: google/gemma-2-2b-it donor into IlyaGusev/gemma-2-2b-it-abliterated.
- Behavior gate: restore harmful refusal while preserving benign helpfulness.
- Current checkpoint tests decoded SAE completeness, not yet individual feature-level explanation.

Causal Target

Patch	Harmful clean	Unsafe continuation	Benign helpful
Abliterated recipient	0.000	0.000	1.000
Full post-FF 16–20	0.750	0.000	1.000
Full post-FF 12–20	1.000	0.000	1.000

The successful activation site is the post-feedforward MLP update over layers 12–20. This aligns the causal target with GemmaScope MLP SAEs.

Sparse-Basis Gate



- Positive: GemmaScope MLP-SAE decoded 12–20 patch reaches harmful clean refusal 1.000 and benign helpfulness 1.000.
- Negative control: corrected layer-16 GemmaScope transcoder reconstruction is weaker, so MLP SAE is the priority sparse branch.
- Caveat: `top_neuron_k1536` also passes on this small screen, so feature selection is still required for novelty.
- Scientific move: go from full decoded reconstruction to sparse feature subsets with matched random controls.

Next Step

- Run donor-active and donor-recipient feature-delta patching inside layers 12–20.
- Compare feature subsets against top-coordinate and random active-feature baselines.
- Expand to held-out harmful prompts and adversarial benign controls before claiming mechanism.
- If subsets fail, the honest claim is behavioral completeness without sparse feature-level explanation.

- `stage2/results/gemma2_2b_dynamic_activation_patch_generation_post_ff/`
- `stage3/results/gemma2_2b_gemmascope_mlp_sae_validation_12_20_generation/`
- `stage3/scripts/validate_gemma2_2b_gemmascope_mlp_sae.py`
- `stage3/results/PROVENANCE.md`